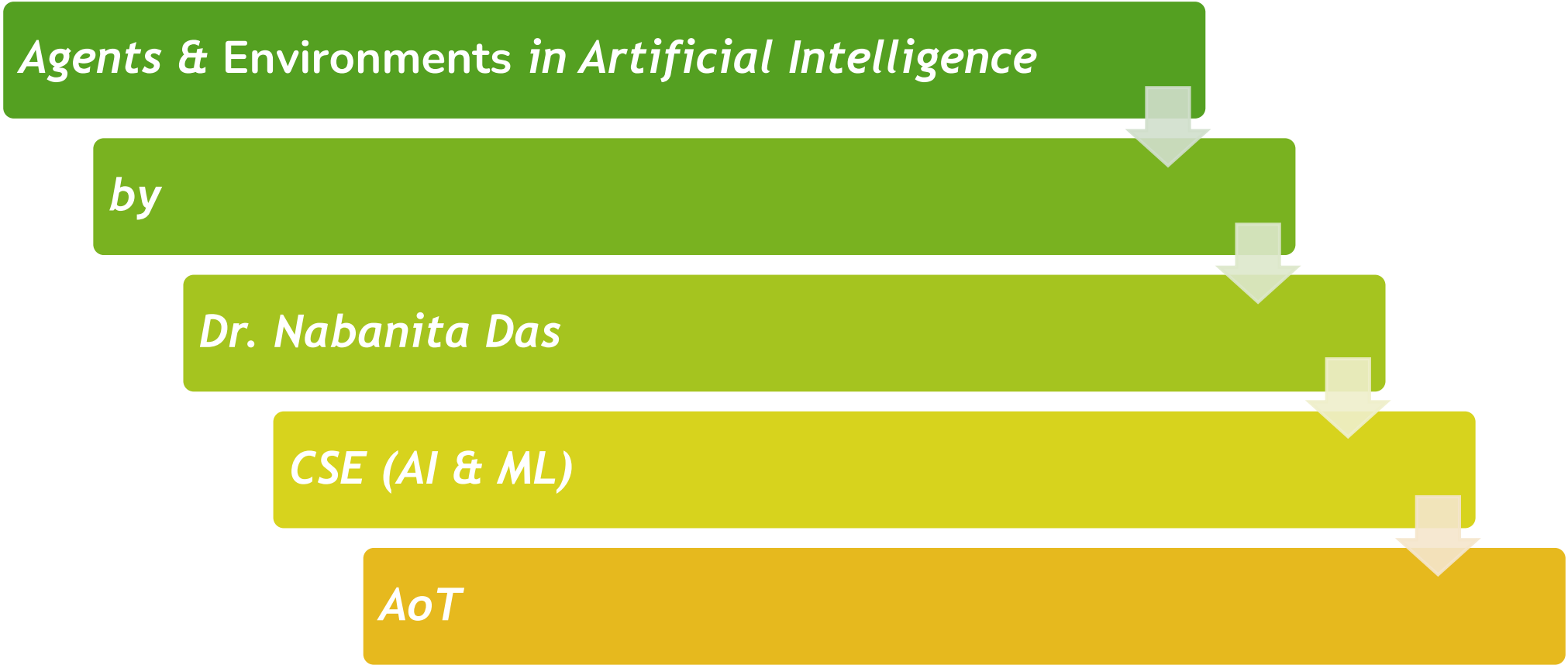


Agents & Environments in Artificial Intelligence



```
graph TD; A[Agents & Environments in Artificial Intelligence] --> B[by]; B --> C[Dr. Nabanita Das]; C --> D[CSE (AI & ML)]; D --> E[AoT];
```

by

Dr. Nabanita Das

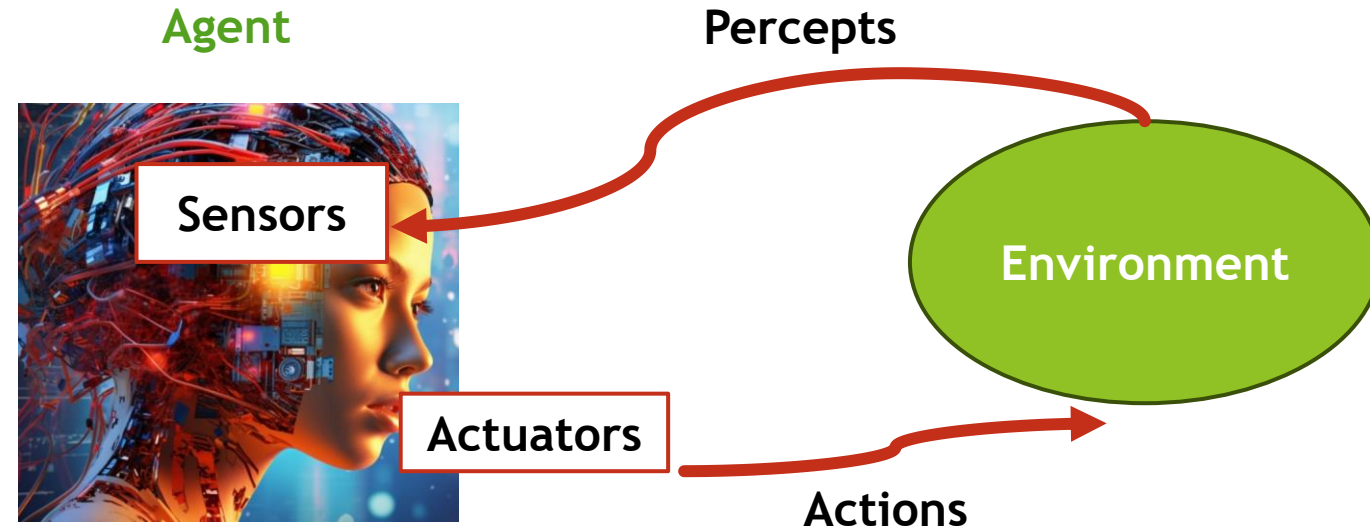
CSE (AI & ML)

AoT

Agents in Artificial Intelligence

► An agent is a computer program or system that is designed to

- ❖ perceive its environment
- ❖ make decisions
- ❖ take actions to achieve a specific goal or set of goals.



- *Perceiving its environment through sensors*
- *Acting upon that environment through actuators*

A Human-agent has eyes, ears, and other organs which act as sensors, and hands, legs, mouth, and other body parts act as actuators.

Structure of an AI Agent

- ▶ **Architecture** is the machinery that the agent executes on. It is a device with sensors and actuators,
 - ▶ for example, a robotic car, a camera, and a PC.
- ▶ **An agent program** is an implementation of an agent function.
- ▶ An **agent function** is a map from the percept sequence(history of all that an agent has perceived to date) to an action

$$\textit{Agent} = \textit{Architecture} + \textit{Agent Program}$$

- ▶ There are many examples of agents in artificial intelligence. Here are a few:
 - ❖ **Intelligent personal assistants:** Siri, Alexa, and Google Assistant.
 - ❖ **Autonomous robots:** Roomba vacuum cleaner and the Amazon delivery robot.
 - ❖ **Traffic management agents:** They can monitor traffic patterns, adjust traffic lights, and reroute vehicles to minimize congestion. Examples of traffic management agents include those used in smart cities around the world.

Types of Agents

- ▶ Agents can be grouped into five classes based on their degree of perceived intelligence and capability :
 - *Simple Reflex Agents*
 - *Model-Based Reflex Agents*
 - *Goal-Based Agents*
 - *Utility-Based Agents*
 - *Learning Agent*

Simple Reflex Agents

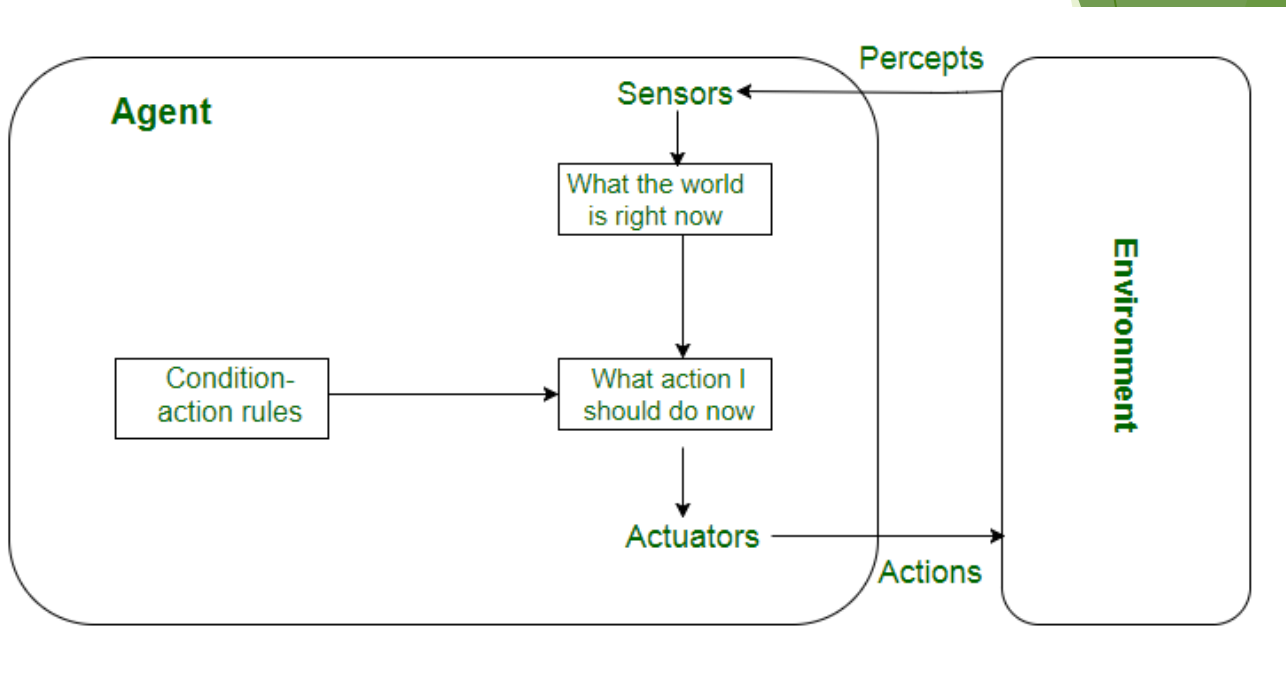
- ▶ ignores the rest of the percept history and act only on the basis of the **current percept**.

- ▶ The agent function is based on the **condition-action rule**.

- ▶ This agent function only succeeds when the environment is fully observable.

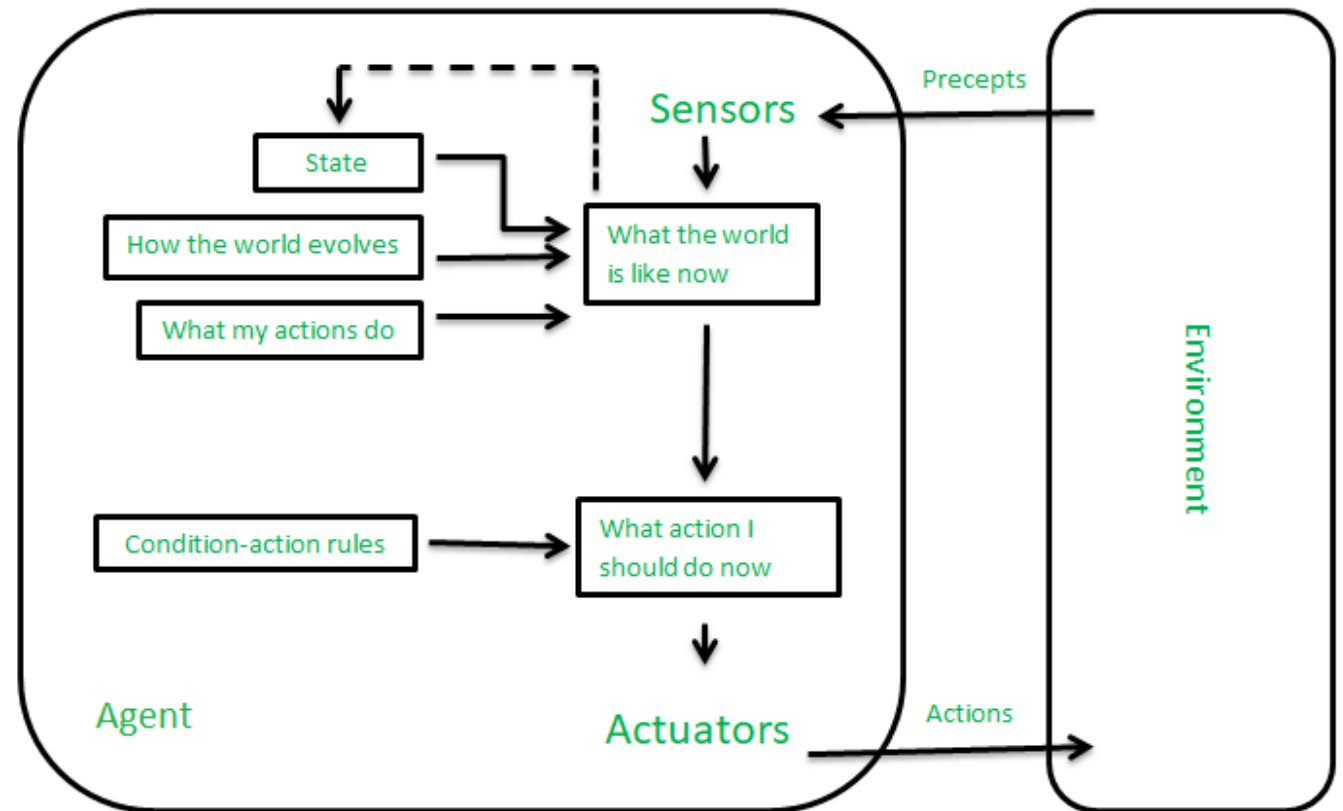
- ▶ **Problems with Simple reflex agents are :**

- ❑ Very limited intelligence.
- ❑ No knowledge of non-perceptual parts of the state.
- ❑ If there occurs any change in the environment, then the collection of rules needs to be updated.



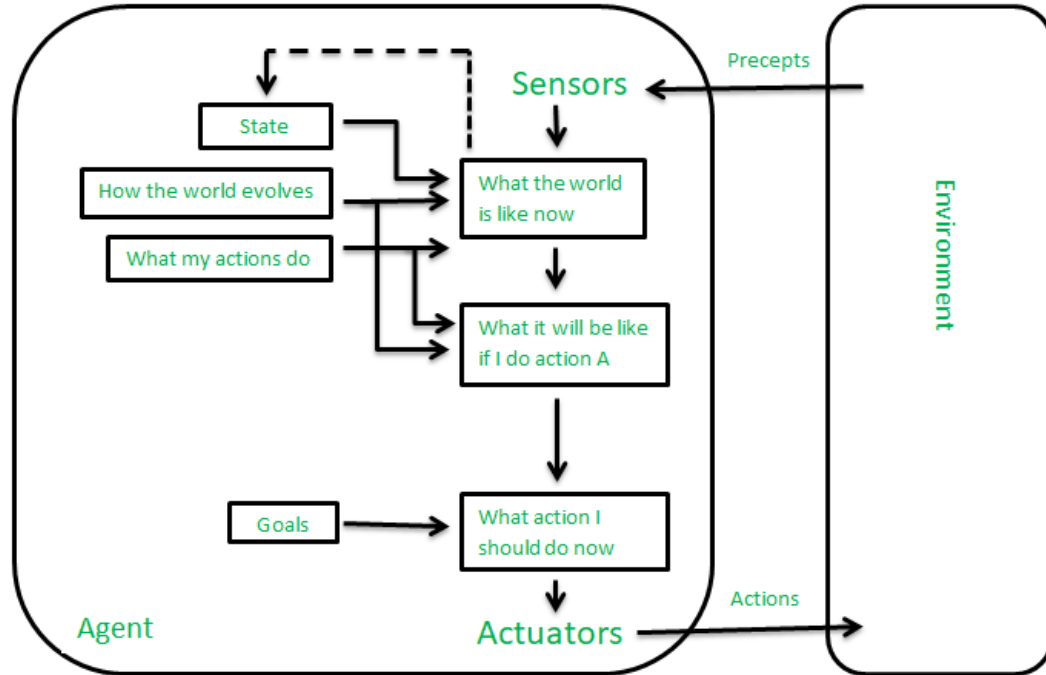
Model-Based Reflex Agents

- The Model-based agent can work in a partially observable environment, and track the situation.
- two important factors:
 - **Model:** It is knowledge about "how things happen in the world," so it is called a Model-based agent.
 - **Internal State:** based on percept history.
- These agents have the model, "which is knowledge of the world" and based on the model they perform actions.



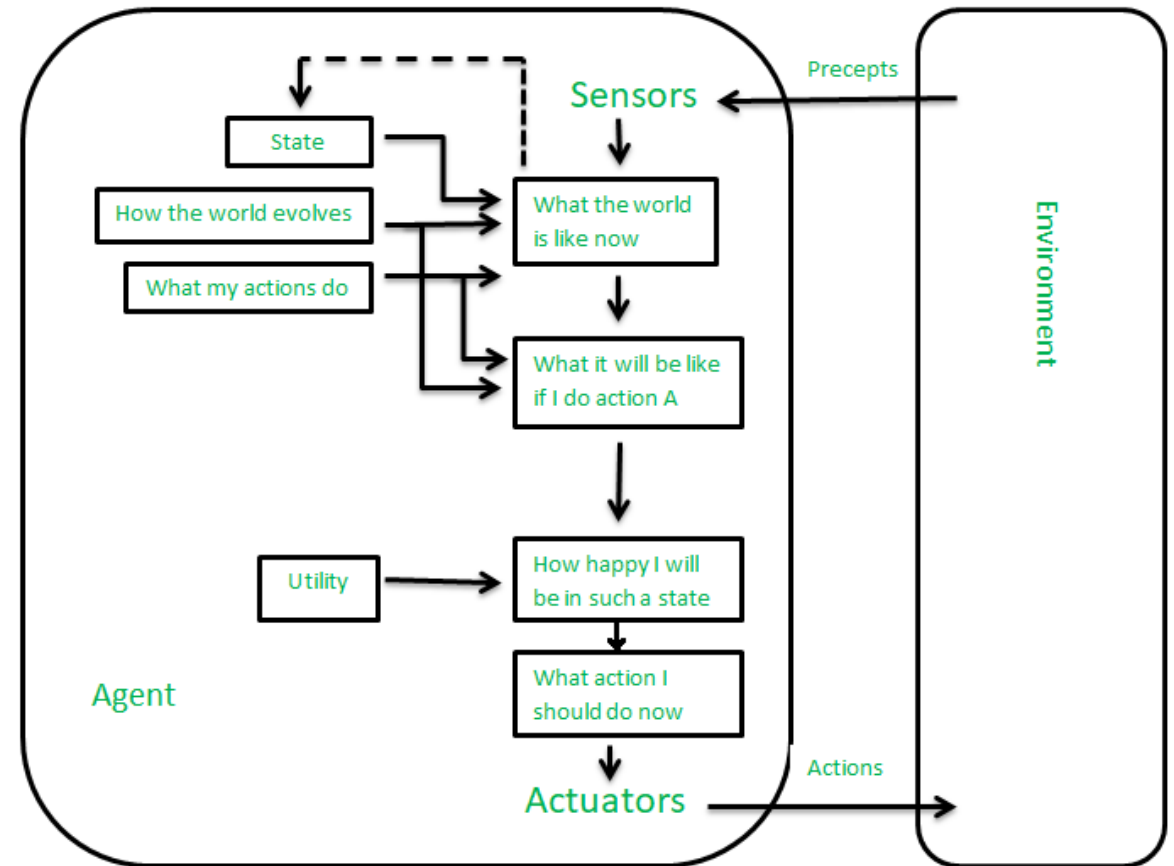
Goal-based agents

- The knowledge of the current state environment is not always sufficient to decide for an agent to what to do.
- The agent needs to know its goal which describes desirable situations.
- Goal-based agents expand the capabilities of the model-based agent by having the "goal" information.
- They choose an action, so that they can achieve the goal.
- These agents may have to consider a long sequence of possible actions before deciding whether the goal is achieved or not. Such considerations of different scenario are called **searching and planning**, which makes an agent proactive.



Utility-based agents

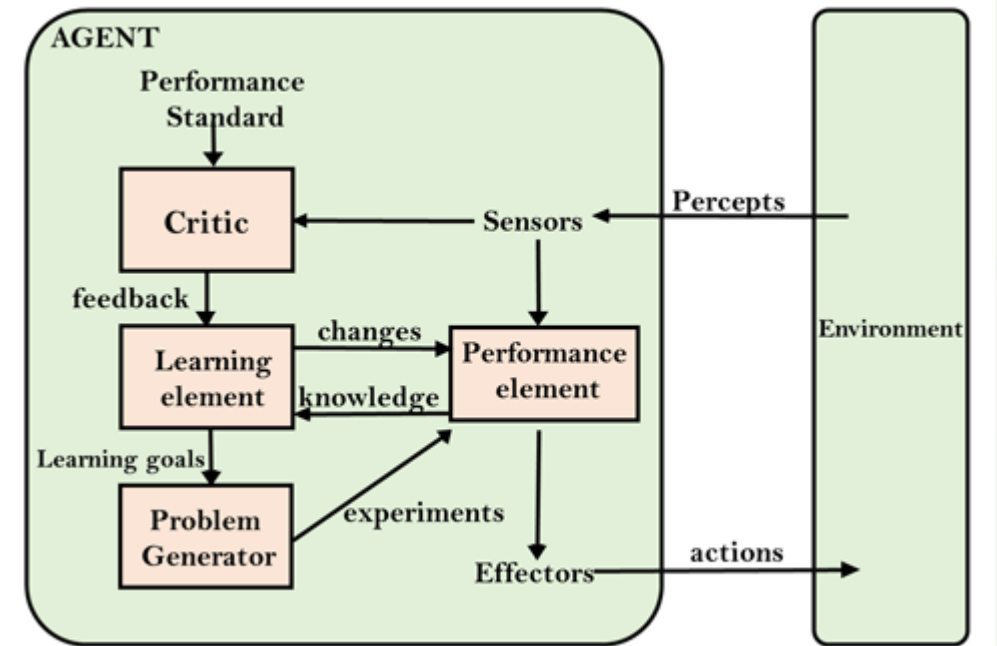
- ❖ Utility-based agent act based not only goals but also the best way to achieve the goal.
- ❖ The Utility-based agent is useful when there are multiple possible alternatives, and an agent has to choose in order to perform the best action.
- ❖ The utility function maps each state to a real number to check how efficiently each action achieves the goals.



Learning Agents

- A learning agent in AI is the type of agent which can learn from its past experiences, or it has learning capabilities.
- It starts to act with basic knowledge and then able to act and adapt automatically through learning.
- A learning agent has mainly four conceptual components, which are:
 - **Learning element:** It is the element which enable agent to learn from environment (**when to do what**)
 - **Critic:** Learning element takes feedback from critic which describes that how well the agent is doing with respect to a fixed performance standard.
 - **Performance element:** It is responsible for selecting external action (**How to do everything**)
 - **Problem generator:** This component is responsible for suggesting actions that will lead to new and informative experiences.

Hence, learning agents are able to learn, analyze performance, and look for new ways to improve the performance.



Multi-Agent Systems

- ▶ A multi-agent system (MAS) is a system composed of multiple interacting agents that are designed to work together to achieve a common goal.

Hierarchical Agents

- ▶ These agents are organized into a hierarchy, with high-level agents overseeing the behavior of lower-level agents. The high-level agents provide goals and constraints, while the low-level agents carry out specific tasks. Hierarchical agents are useful in complex environments with many tasks and sub-tasks.

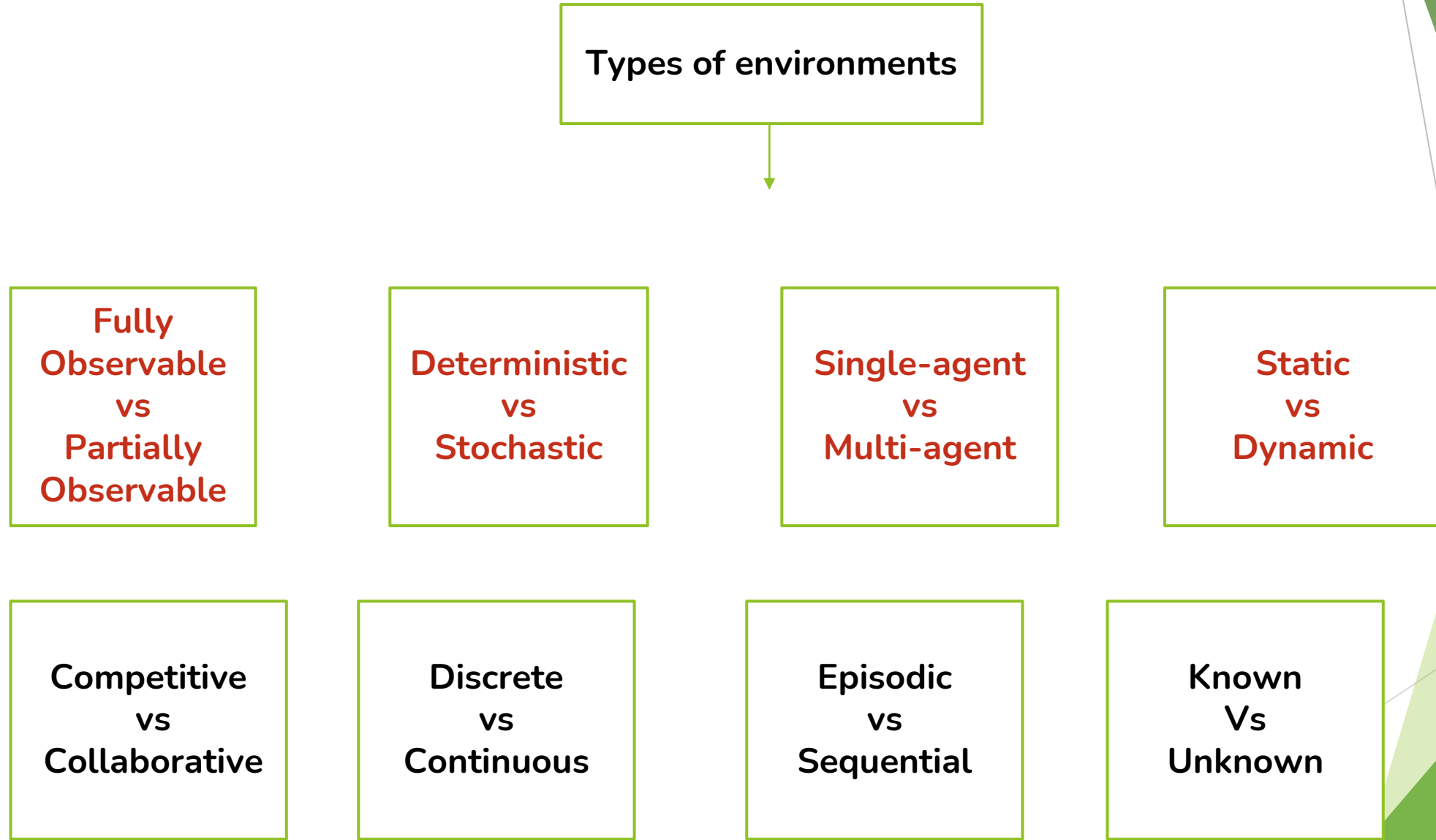
PEAS in Artificial Intelligence

- ▶ PEAS stands for a
 - ❖ *Performance measure*
 - ❖ *Environment*
 - ❖ *Actuator*
 - ❖ *Sensor*
- ▶ **Performance Measure:** Performance measure is the unit to define the success of an agent. Performance varies with agents based on their different precepts.
- ▶ **Environment:** Environment is the surrounding of an agent at every instant. It keeps changing with time if the agent is set in motion.
- ▶ **Actuator:** An actuator is a part of the agent that delivers the output of action to the environment.
- ▶ **Sensor:** Sensors are the receptive parts of an agent that takes in the input for the agent.

Agent	Performance Measure	Environment	Actuator	Sensor
<i>Hospital Management System</i>	Patient's health, Admission process, Payment	Hospital, Doctors, Patients	Prescription, Diagnosis, Scan report	Symptoms, Patient's response
<i>Automated Car Drive</i>	The comfortable trip, Safety, Maximum Distance	Roads, Traffic, Vehicles	Steering wheel, Accelerator, Brake, Mirror	Camera, GPS, Odometer
<i>Subject Tutoring</i>	Maximize scores, Improvement is students	Classroom, Desk, Chair, Board, Staff, Students	Smart displays, Corrections	Eyes, Ears, Notebooks
<i>Part-picking robot</i>	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arms and hand	Camera, joint angle sensors
<i>Satellite image analysis system</i>	Correct image categorization	Downlink from orbiting satellite	Display categorization of scene	Color pixel arrays

Types of Environments in AI

Types of environments



```
graph TD; A[Types of environments] --> B[Fully Observable vs Partially Observable]; A --> C[Deterministic vs Stochastic]; A --> D[Single-agent vs Multi-agent]; A --> E[Static vs Dynamic]; A --> F[Competitive vs Collaborative]; A --> G[Discrete vs Continuous]; A --> H[Episodic vs Sequential]; A --> I[Known Vs Unknown];
```

**Fully
Observable
vs
Partially
Observable**

**Deterministic
vs
Stochastic**

**Single-agent
vs
Multi-agent**

**Static
vs
Dynamic**

**Competitive
vs
Collaborative**

**Discrete
vs
Continuous**

**Episodic
vs
Sequential**

**Known
Vs
Unknown**

Fully Observable vs Partially Observable

Aspects	Fully Observable Environment	Partially Observable Environment
Access	Complete access to the environment's state	Limited or incomplete access to environment's state
Information Availability	All relevant aspects are directly observable	Some aspects may be obscured, uncertain, or missing
Decision-Making	Straightforward decision-making based on complete information	More complex due to incomplete information
Memory Requirement	No or minimal memory requirements.	Needed to track previous observation.
Solution	Optimal and Transparent.	Sub-optimal and unexpected.
Example	Chess	Autonomous driving, Robot navigation

Deterministic vs Stochastic

- ❖ When a uniqueness in the agent's current state completely determines the next state of the agent, the environment is said to be deterministic.
- ❖ The stochastic environment is random in nature which is not unique and cannot be completely determined by the agent.
- ❖ **Examples:**
 - ❖ **Chess** – there would be only a few possible moves for a chess piece at the current state and these moves can be determined.
 - ❖ **Self-Driving Cars**- the actions of a self-driving car are not unique, it varies time to time.

Single-agent vs Multi-agent

- ❖ An environment consisting of only one agent is said to be a single-agent environment.
 - ❖ A person left alone in a maze is an example of the single-agent system.
- ❖ An environment involving more than one agent is a multi-agent environment.
 - ❖ The game of football is multi-agent as it involves 11 players in each team.

Dynamic vs Static

- ❖ An environment that keeps constantly changing itself when the agent is up with some action is said to be dynamic.
 - ❖ A roller coaster ride is dynamic as it is set in motion and the environment keeps changing every instant.
- ❖ An idle environment with no change in its state is called a static environment.
 - ❖ An empty house is static as there's no change in the surroundings when an agent enters.

Competitive vs Collaborative

- ❖ An agent is said to be in a competitive environment when it competes against another agent to optimize the output.
 - ❖ The game of chess is competitive as the agents compete with each other to win the game which is the output.
- ❖ An agent is said to be in a collaborative environment when multiple agents cooperate to produce the desired output.
 - ❖ When multiple self-driving cars are found on the roads, they cooperate with each other to avoid collisions and reach their destination which is the output desired.

Discrete vs Continuous

- ❖ If an environment consists of a finite number of actions that can be deliberated in the environment to obtain the output, it is said to be a discrete environment.
 - ❖ The game of chess is discrete as it has only a finite number of moves. The number of moves might vary with every game, but still, it's finite.
- ❖ The environment in which the actions are performed cannot be numbered i.e. is not discrete, is said to be continuous.
 - ❖ Self-driving cars are an example of continuous environments as their actions are driving, parking, etc. which cannot be numbered.

Known vs Unknown

- ❖ In a known environment, the output for all probable actions is given.
- ❖ Obviously, in case of unknown environment, for an agent to make a decision, it has to gain knowledge about how the environment works.

Episodic vs Sequential

- ❖ In an Episodic task environment, each of the agent's actions is divided into atomic incidents or episodes. There is no dependency between current and previous incidents. In each incident, an agent receives input from the environment and then performs the corresponding action.
 - ❖ Example: Consider an example of Pick and Place robot, which is used to detect defective parts from the conveyor belts. Here, every time robot(agent) will make the decision on the current part i.e. there is no dependency between current and previous decisions.
- ❖ In a Sequential environment, the previous decisions can affect all future decisions. The next action of the agent depends on what action he has taken previously and what action he is supposed to take in the future.
 - ❖ Example:
Checkers- Where the previous move can affect all the following moves.